Own data collection guide

What to record, how to record it, the exact file names, and what happens next

1. Why we need it

The models learned from public internet photos. They are weakest on **our** room, light, cameras and PPE: gloves, safety shoes and dim light. 30-40 seconds of video per scene from our own cameras fixes exactly that. Every video becomes 30-40 training frames. **24 short videos (about 15 minutes of filming) is enough.**

The two most important videos are the shoe videos (ppe_14 and ppe_15): safety-shoe alerts only switch on after they exist.

2. Before you start

- **Cameras:** record with the cameras we demo with - the phone (normal camera app, 1080p, landscape), the laptop camera and the overhead USB webcam (Windows **Camera** app; switch camera with the camera-switch button). Mix them: about one third of the videos per camera. The overhead webcam scenes are the CCTV view.
- **People:** 2-3 different people, different clothes. Everyone agrees to be filmed.
- **Kit:** helmet, hi-vis vest, gloves, safety goggles, safety shoes (the real ones we demo with).
- **Length:** 30-40 seconds per video. Keep the camera still (tripod, shelf or tape).
- **Movement:** walk slowly, turn around, raise and lower hands, come close (1.5 m) and go far (5 m).
- **Whole body** in the picture whenever shoes matter (head to feet).

3. Where the files go and how to name them

- Copy PPE videos to `sentineleye\datasets\raw_videos\ppe\` and fire/smoke videos to `sentineleye\datasets\raw_videos\fire\`.
- Rename each file **exactly** as in the tables (lower case, underscores, .mp4). The name tells the software what is in the video.
- Videos whose name starts with neg_ contain **no fire and no smoke**. They need no labelling and teach the model what is NOT fire.
- A video went wrong? Record it again with the same name and replace the file.

4. PPE videos (14 must-have)

Done	File name	What to show	Camera / light
☐	ppe_01_fullkit_day.mp4	All PPE on: helmet, vest, gloves, goggles, safety shoes. 1-3 people	phone, normal light
☐	ppe_02_fullkit_dim.mp4	Same as 01	laptop, one dim lamp only
☐	ppe_03_helmet_off.mp4	Full kit, NO helmet	overhead webcam
☐	ppe_04_gloves_off.mp4	Full kit, NO gloves, bare hands clearly visible and moving	laptop
☐	ppe_05_goggles_off.mp4	Full kit, NO goggles, face to the camera	laptop
☐	ppe_06_vest_off.mp4	Full kit, NO vest (normal shirt)	phone
☐	ppe_07_no_ppe.mp4	Normal clothes, no PPE at all	overhead webcam
☐	ppe_08_cap_shirt_lookalikes.mp4	Cap, hood, scarf; orange / yellow / white T-shirt; normal spectacles, sunglasses. No real PPE	phone
☐	ppe_14_footwear_normal.mp4	**Sneakers, sandals, slippers, formal shoes, bare feet.** Close-up of the feet AND full body, walking	phone, normal light
☐	ppe_15_footwear_safety.mp4	**The safety shoes** from all sides, close-up AND full body, walking	phone, normal light
☐	ppe_17_two_in_line.mp4	Two people: one behind the other, then side by side, then crossing. One in full kit, one without	overhead webcam
☐	ppe_20_overhead_cctv.mp4	2-3 people walking around below the overhead camera, mixed PPE	overhead webcam
☐	ppe_21_far.mp4	Workers 5-8 m away, full kit and without	phone
☐	ppe_22_backlit.mp4	Standing in front of a bright window	laptop

Optional if time allows: ppe_09_all_off_dim (no PPE, dim light), ppe_10_helmet_in_hand (helmet carried, not worn), ppe_19_group_mixed (3-4 people overlapping), ppe_24_warehouse_* (real workshop or warehouse footage, if you are allowed to film there).

5. Fire and smoke videos (10 must-have)

Safety: only candle, lighter, match, incense, or paper in a metal tray. Outdoors or next to an open window, water bucket or extinguisher next to you, an adult supervising. Never near curtains, paper or anything that can catch fire.

Done	File name	What to show
☐	fire_01_candle.mp4	Candle flame, near (1 m) and far (3 m), moving the camera slowly
☐	fire_02_lighter.mp4	Lighter / match flame, several times on and off
☐	fire_04_paper_tray.mp4	Paper burning in a metal tray outdoors: starts small, grows, dies down
☐	fire_05_agarbatti.mp4	Incense smoke close and far; also smoke right after blowing out a candle
☐	neg_01_dark_room.mp4	Lights off or one dim lamp, people moving, 60 seconds
☐	neg_04_white_tshirt.mp4	White T-shirt, faces and hair close to the camera
☐	neg_06_phone_flash.mp4	Phone torch / flash and desk lamp pointed at the camera, tube light
☐	neg_08_red_cloth.mp4	Red or orange cloth, orange vest, red bag moving in front of the camera
☐	neg_10_kettle_steam.mp4	Steam from a kettle or hot water
☐	neg_13_phone_screen_fire.mp4	A phone or laptop screen playing a fire video (kept for the hardware flame-sensor test)

6. Do and don't

Do	Don't
Keep the camera still; let people move	Wave the camera around
Show each PPE item clearly for a few seconds	Stand in one pose for the whole video
Mix near and far, front and side, light and dim	Record everything from the same spot
Record what fooled the system before (cap, shirt, lamp, steam)	Use real big fires or anything unsafe
Name files exactly as in the tables	Use spaces or capital letters in file names

7. After recording: what happens next

Step	Who	What
1	You	Copy the videos into datasets\raw_videos\ppe\ and \fire\ with the right names
2	You	VS Code terminal (venv on): `python -m tools.frames extract --task ppe` and `python -m tools.frames extract --task fire`. It cuts each video into frames and pre-labels them with the current models
3	You	`python -m tools.shoes crops` then `python -m tools.shoes train` (about 5 minutes on the laptop). Restart: safety-shoe alerts are live
4	Claude	Checks the frames and pre-labels, writes the train / validation / hard-test split
5	You	`python -m tools.frames pack --task ppe` (and fire). Upload the zip to Roboflow and correct the boxes (rules below). Split the scenes between team members
6	You	Export from Roboflow as YOLOv8, then `python -m tools.frames import --task ppe --src <export>`
7	Claude + Colab	Final training run (stage B) with public data + your data; then measured before/after numbers

Labelling rules in Roboflow

- Draw a box on every visible **head, face, foot**, every **bare hand** (a gloved hand is labelled **gloves**, not hand), and every PPE item that is **worn**: helmet, goggles, mask, vest, gloves.
- A helmet held in the hand gets **no** helmet box. Normal spectacles get **no** goggles box. A plain shirt gets **no** vest box.
- Partly hidden but clearly visible = box it. Mostly hidden = leave it.
- Wrong pre-label boxes: delete them. Missing boxes: draw them. Correct boxes: leave them.

Target: all 24 videos recorded by early afternoon, so labelling can finish and the final training can run this evening.